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Introduction

This quick-start guide covers:

- How to set your volume origin with the active wand (*IR version 2*)
- How to set your floor plane
- How to confirm your floor plane is accurate

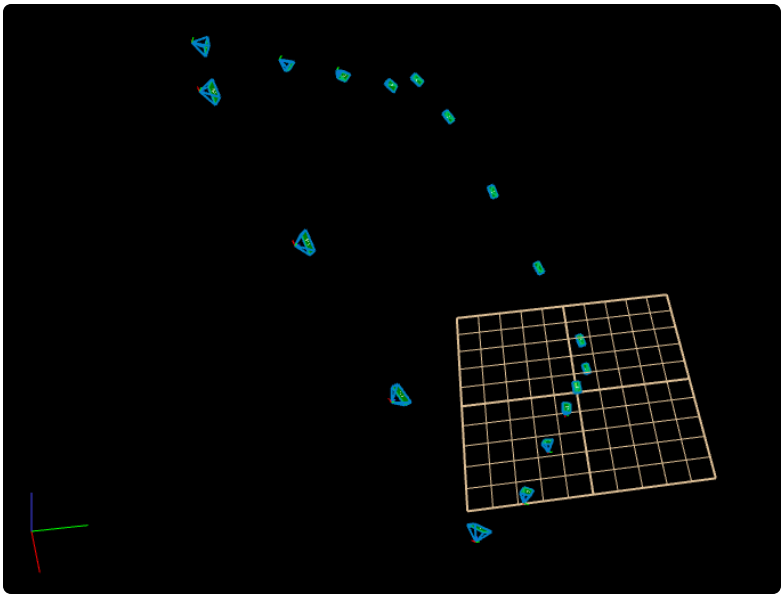
Important - this guide assumes you have already calibrated your cameras.
If you have not yet calibrated your cameras, please go to the calibrating cameras guide and complete the steps there.

This is the third of the three-part quick-start system preparation guide:

1. Masking your Cameras
2. Calibrating your Cameras
3. ***Setting Volume Origin and Plane***

Setting your Volume (Quick-Start Guide)

During the calibrating cameras guide you will have noticed that your cameras are not parallel to the floor, often instead intersecting the floor plane.



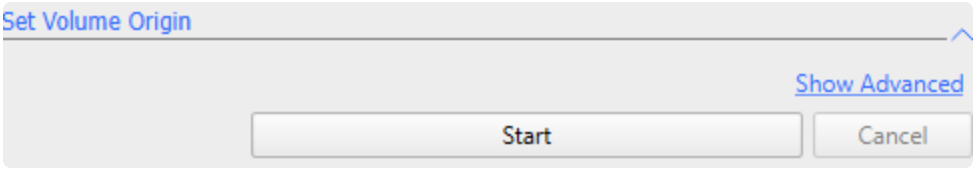
This is because the cameras do not have reference to the origin or floor plane of your volume from calibration, and you will need to set this in order to complete the calibration.

Once you have completed this guide you will be ready to capture trial data.

Setting the Volume Origin

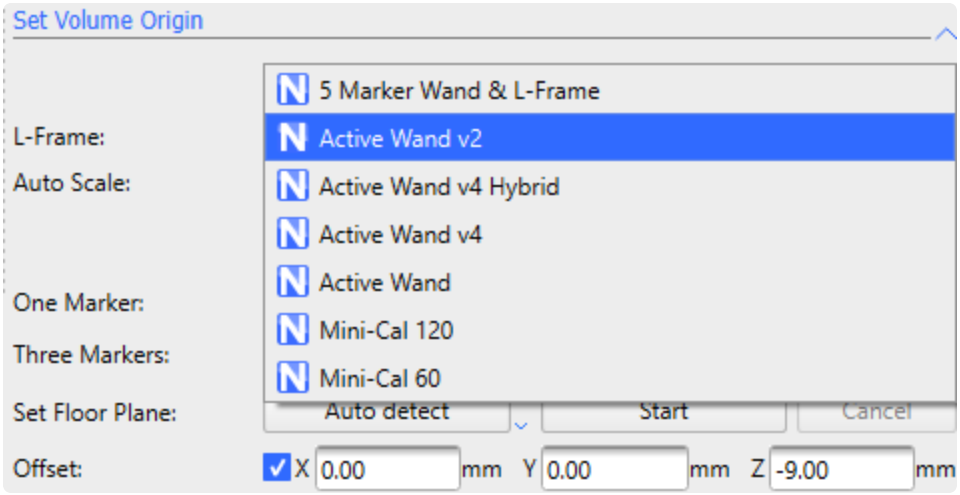
Nexus can determine the origin of the volume using the active wand.

Before setting volume, ensure the advanced options are toggled ('show advanced').



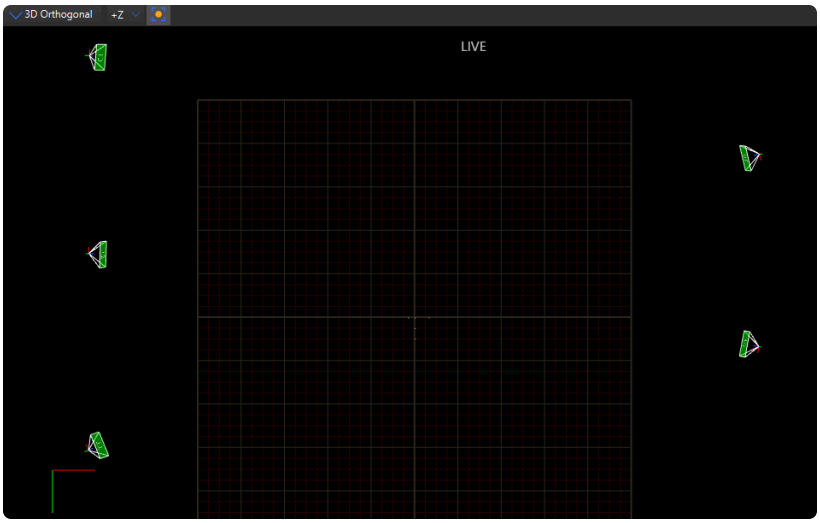
Click 'show advanced' to toggle advanced properties

INB Lab specific: Confirm the L-Frame setting is set to the '**Active Wand v2**'.



Selecting the active wand v2

Place the wand within the centre of your volume. You can use the Nexus 3D Perspective view or 3D Orthogonal view to determine the position of the wand relative to the volume.



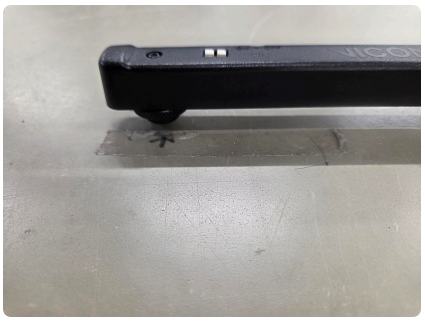
Orthogonal view showing an active wand within the volume

If you regularly use the same origin it is worth considering marking the place in which the wand is placed, to ensure consistent volume origin.

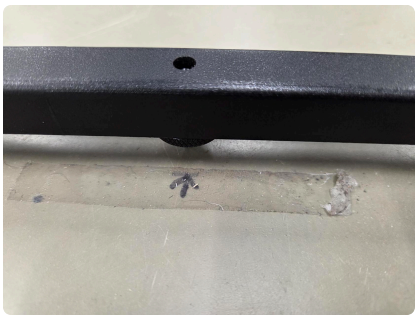
INB-specific

Within the INB lab there are indicators showing the volume origin.

Place the wand so that the middle vertical pivot is against the vertical strip arrow, and the top-left horizontal pivot is against the horizontal strip arrow, using the tape strips as a right-angle guideline for the wand.



Horizontal bar position



Vertical bar position



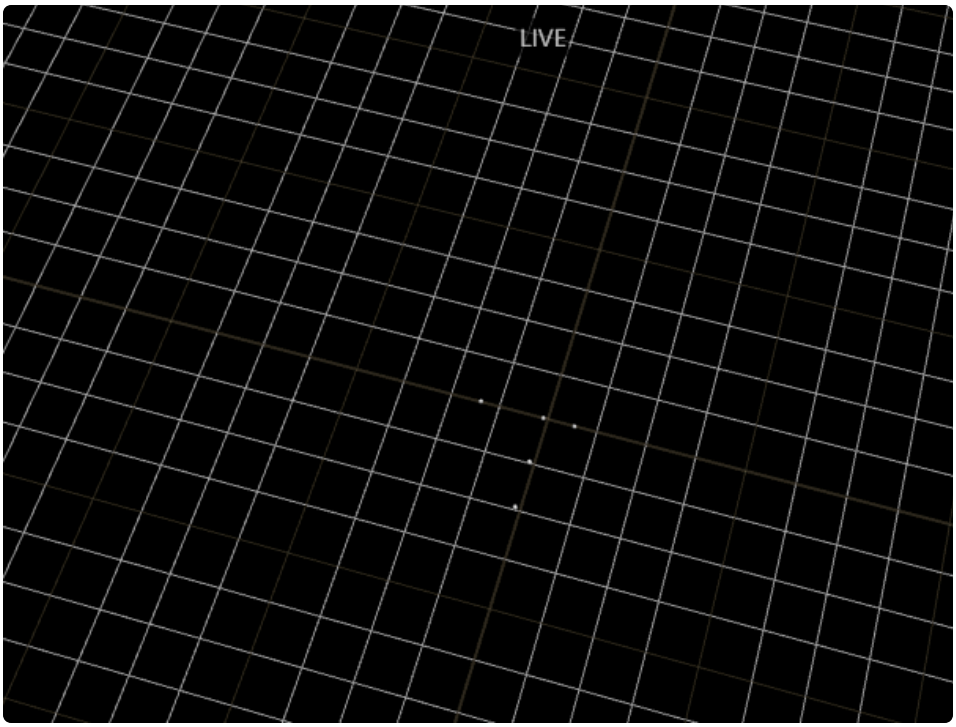
Wand against marker strips

When placing the wand on the floor ensure that the level bubble within the wand is level.



Wand bubble - within the centre level guide

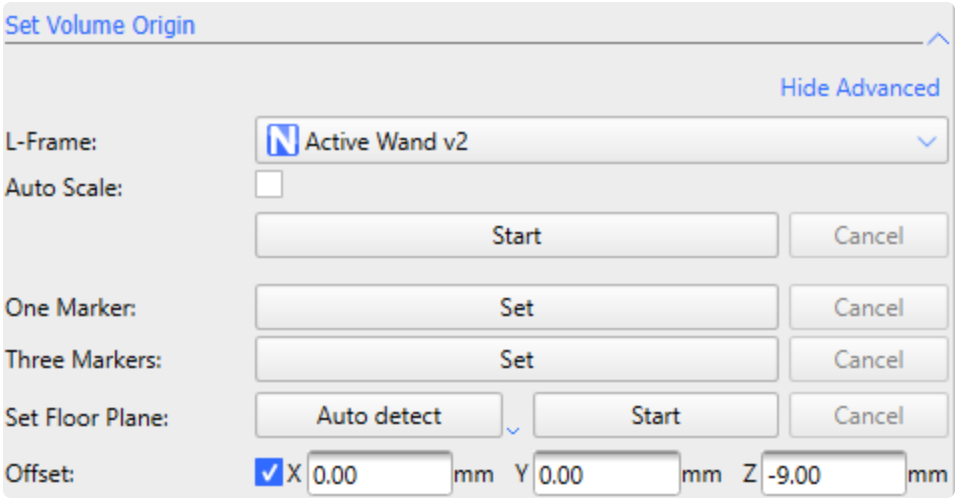
Using the 3D perspective view you can confirm the wand is within the volume.



Wand shown within Nexus 3D perspective view

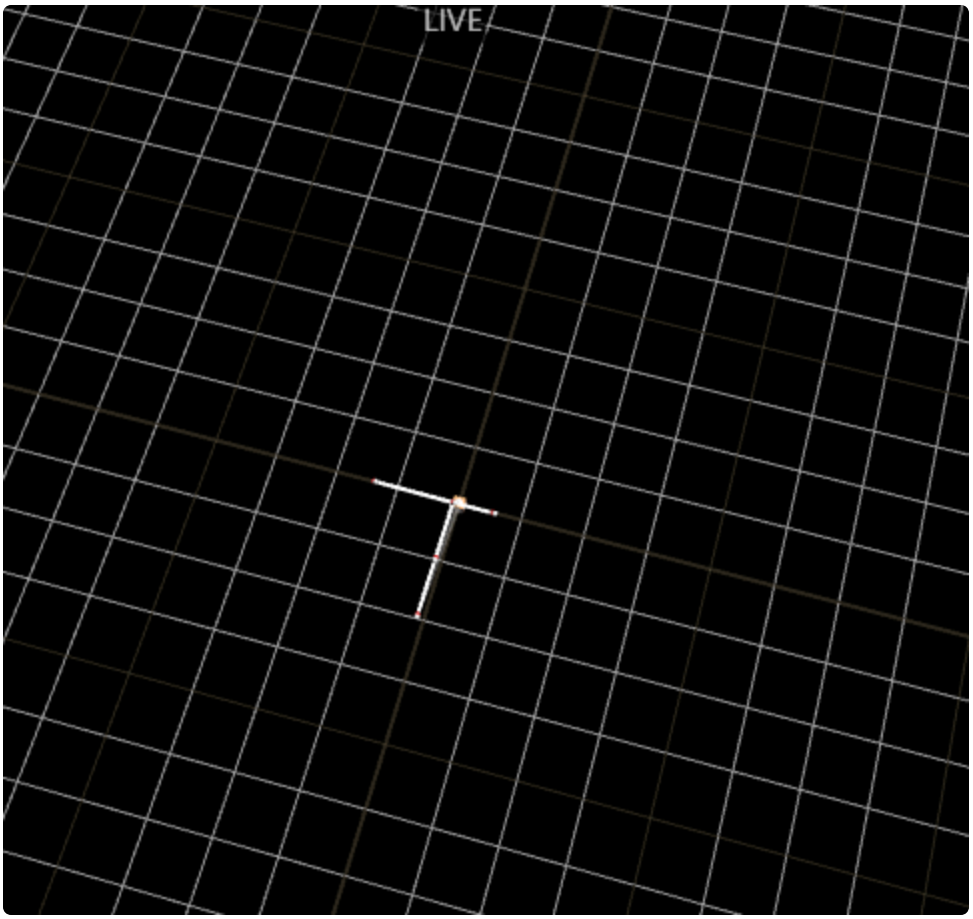
Once the wand is in position you can click '**start**' to set the volume - see the '*set volume origin*' section of the **tools panel**.

If you are using the active wand, do not use the 'auto-scale' setting.



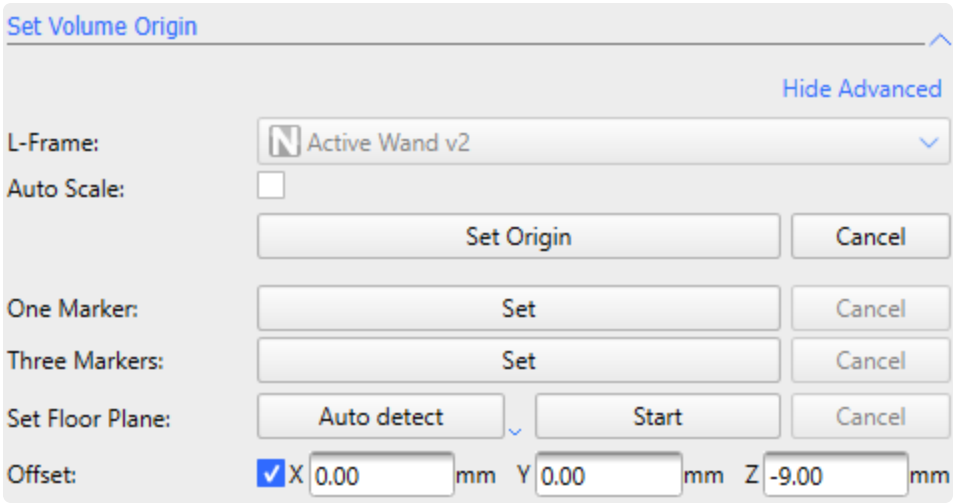
Setting origin - starting view

At this time you will notice Nexus starts labelling the wand, if enough cameras can see it.



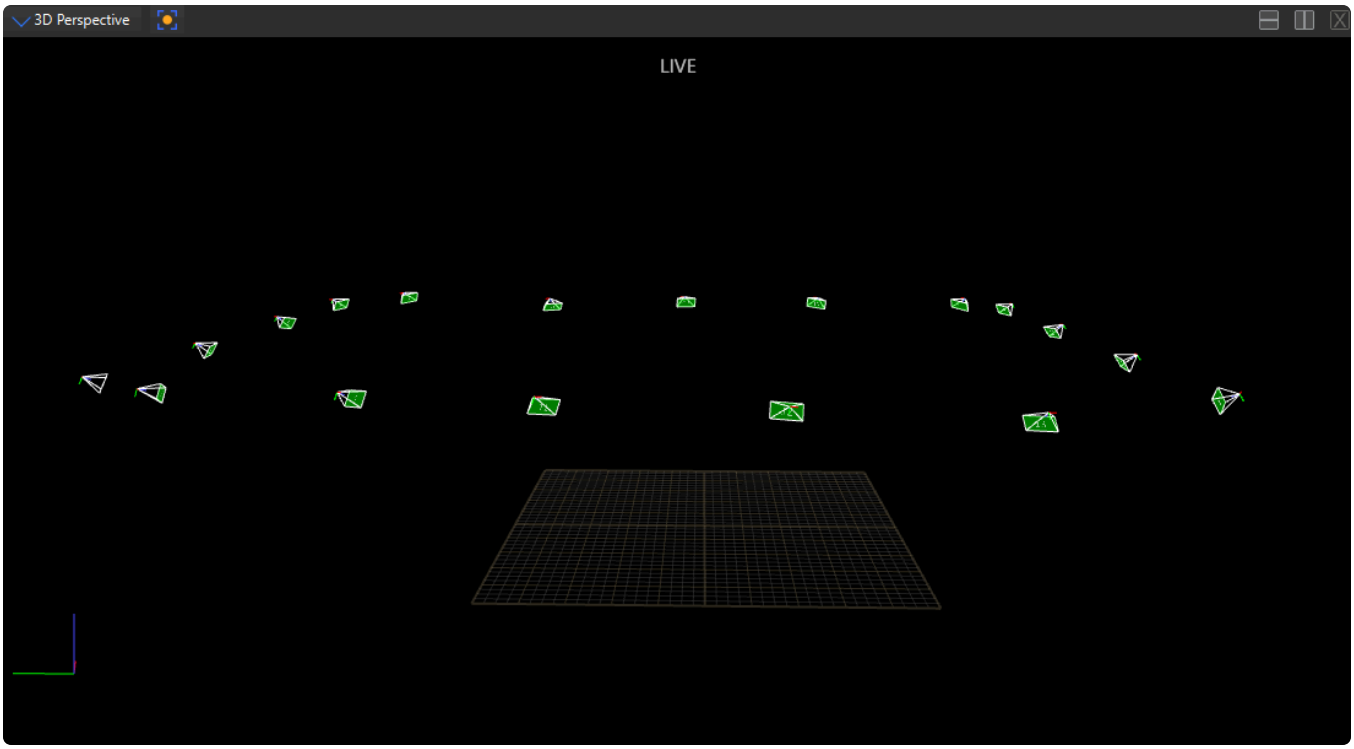
Wand labelled within Nexus 3D perspective view

The '**start**' button will change to say 'set origin'. Click the button again to set the origin.



Setting origin - confirming origin view

If your cameras have successfully detected the volume origin, and are successfully calibrated, the cameras will now be relative to the floor plane as well as each other.



Setting Floor Plane

The last step you need to complete before being ready to capture trial data, is setting the floor plane. In order to set the floor plane you will need to switch off the active wand (as its LEDs are elevated from the floor), and utilise reflective markers

INB lab specific: The most effective tool to set the floor plane are the B&L Engineering Pearl markers provided with the Vicon system.

Place clusters of markers throughout the volume - you should place clusters of 3 markers, spaced out within the volume.

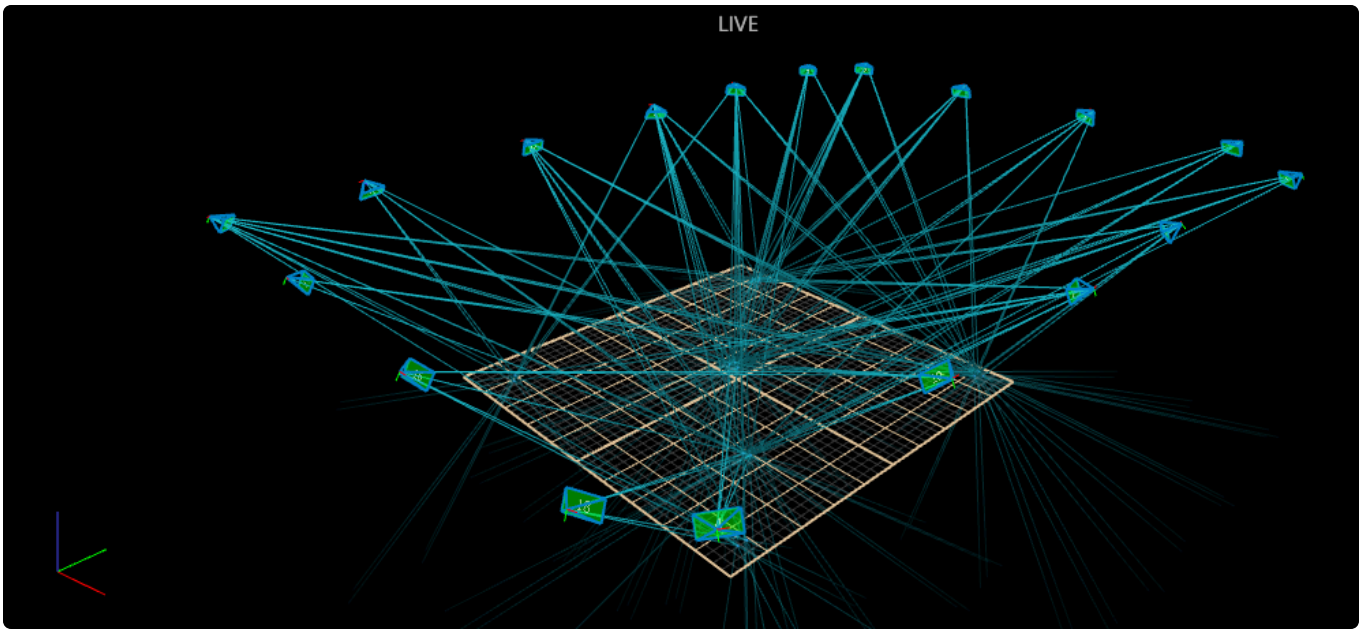
Do not place markers within clusters too far apart, as this can create *'ghost markers'* from reflections, and impact setting floor plane.

TODO add ghost marker error screenshot

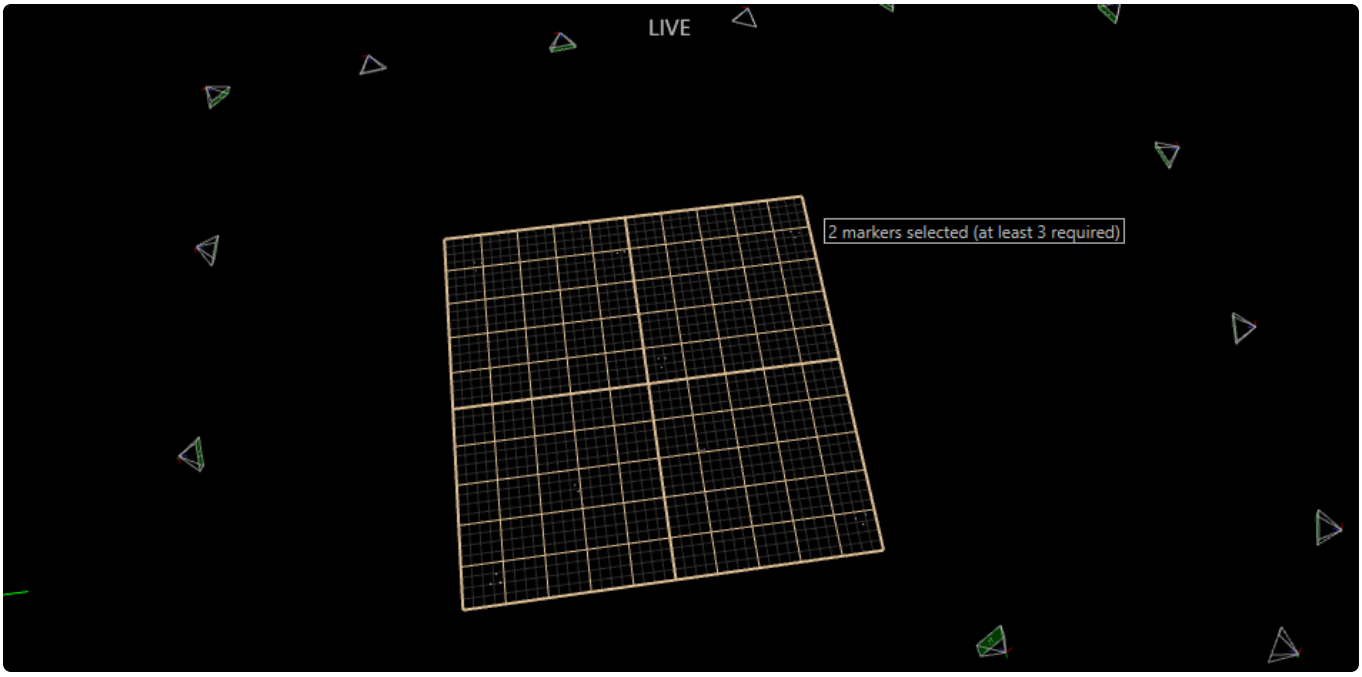
The larger your volume the more clusters you will need - any changes in the level of the floor will be multiplied over distance. Adding additional clusters within the volume when setting floor plane assists Nexus with determining the slope of the floor.

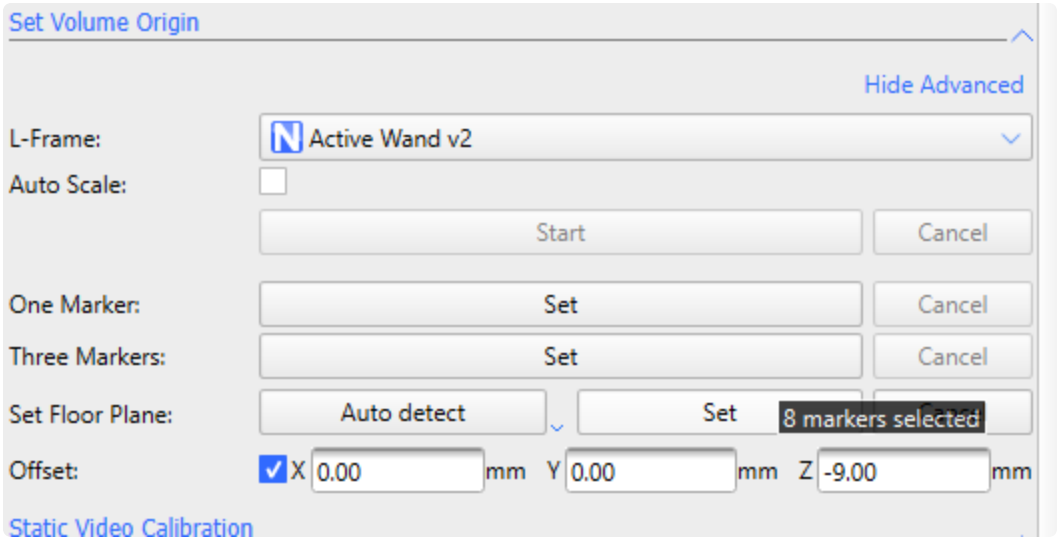
Once you have done so you can click auto-detect for the cameras to start detecting the markers.

Click start and set to confirm setting markers.

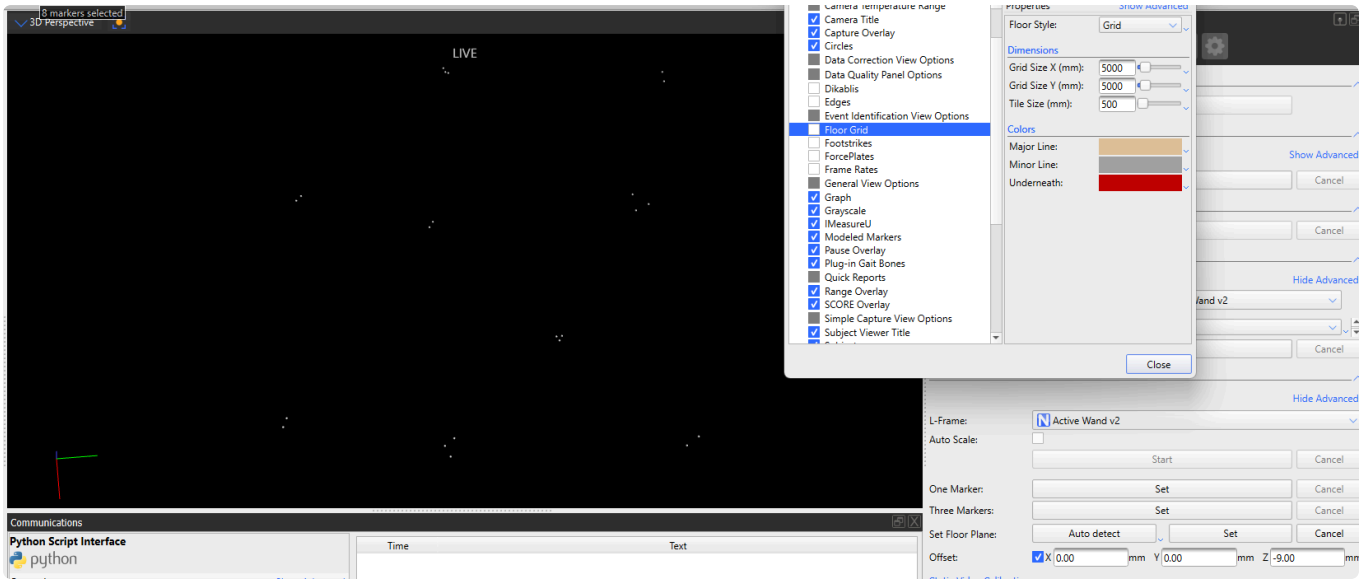


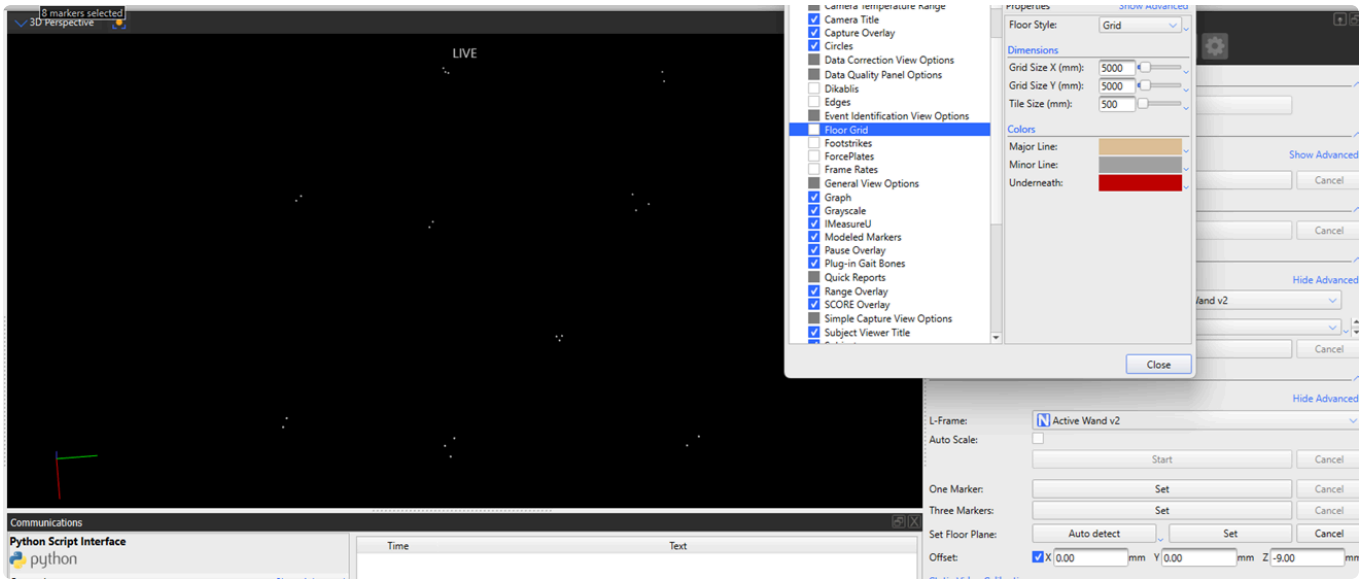
Selecting markers

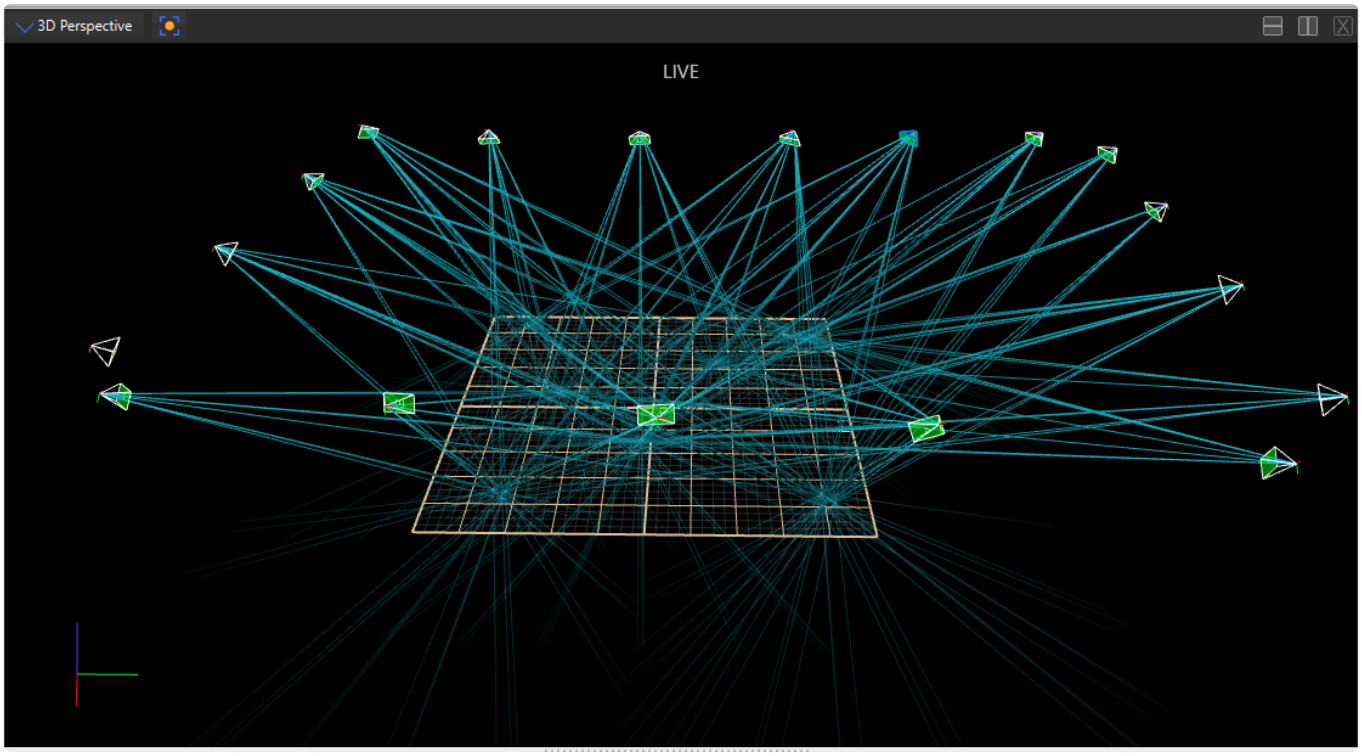


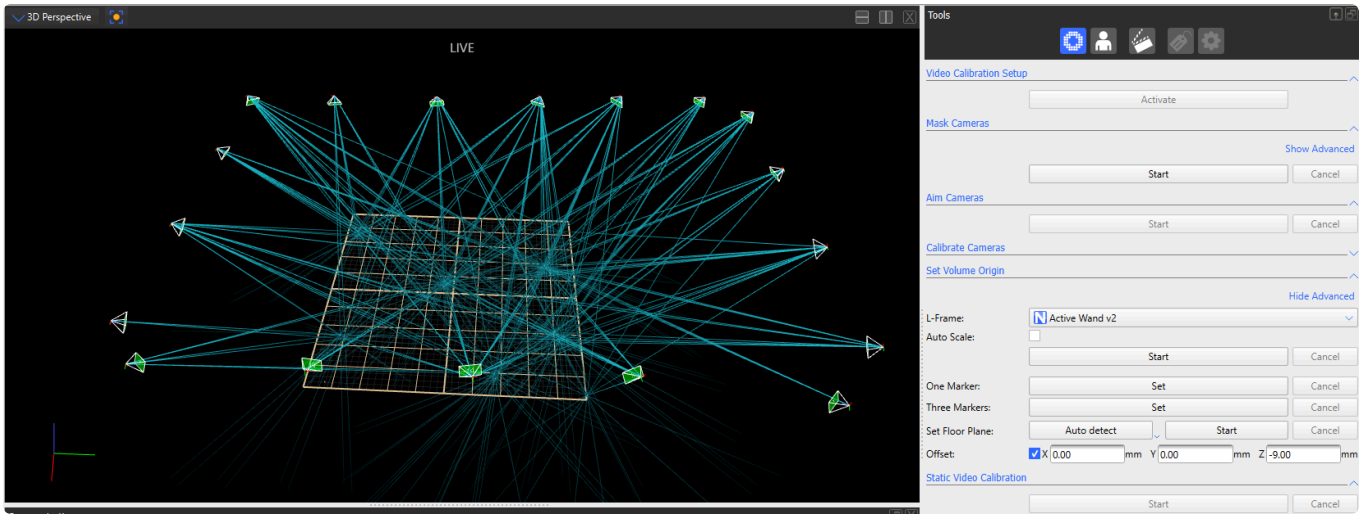


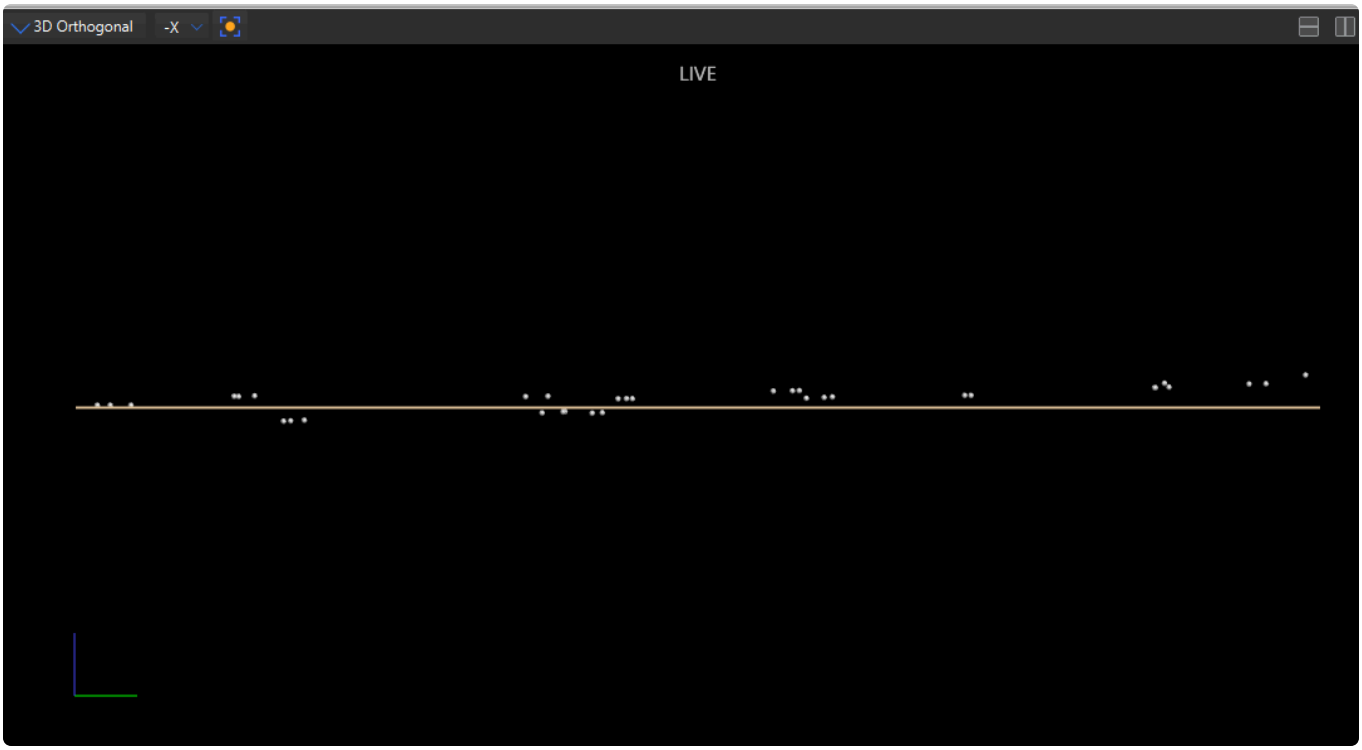
Turning off the floor grid

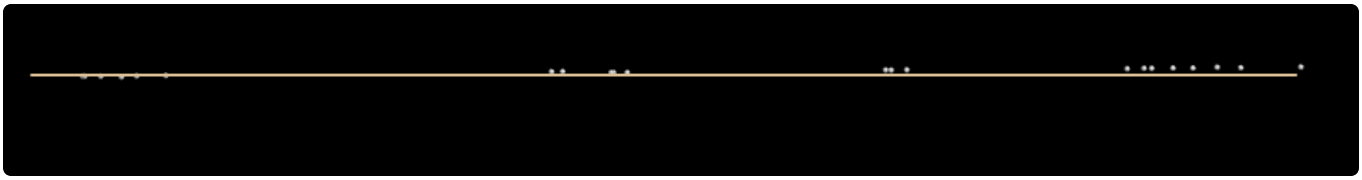


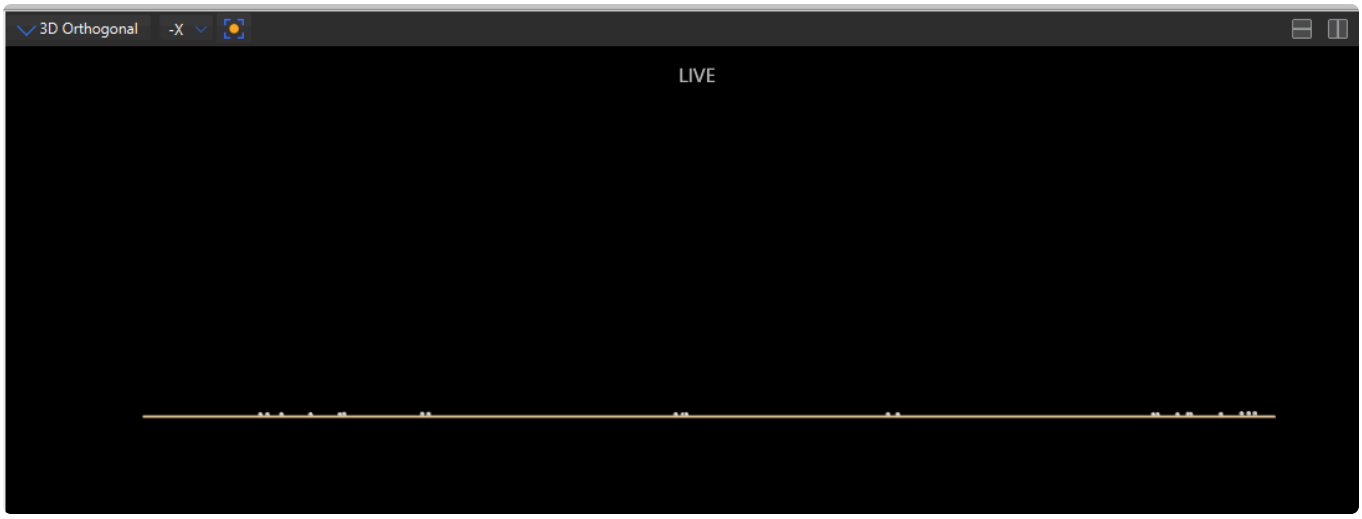



3D Perspective
LIVE

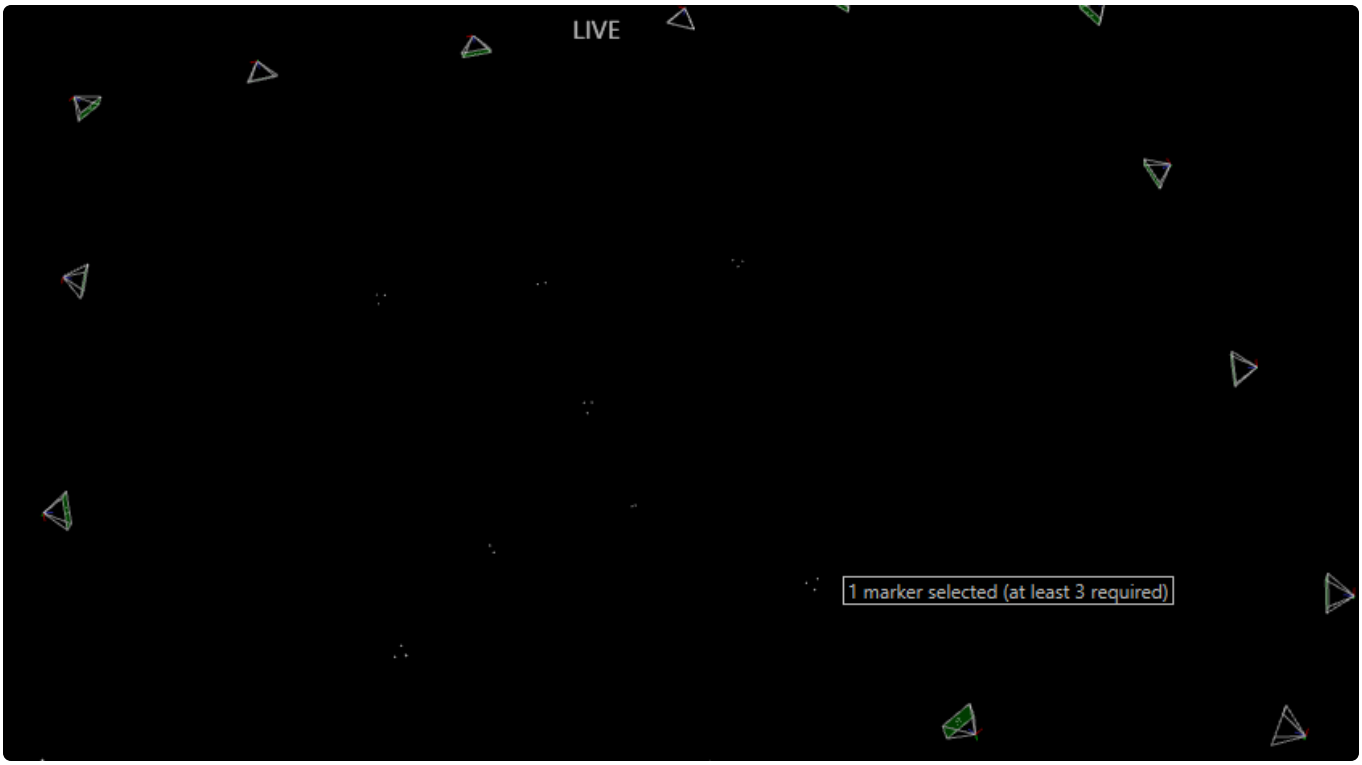

3D Perspective
LIVE
Tools
Video Calibration Setup
Activate
Mask Cameras
Show Advanced
Start
Cancel
Aim Cameras
Start
Cancel
Calibrate Cameras
Set Volume Origin
Hide Advanced
L-Frame:
Active Wand v2
Auto Scale:
Start
Cancel
One Marker:
Set
Cancel
Three Markers:
Set
Cancel
Set Floor Plane:
Auto detect
Start
Cancel
Offset:
X 0.00 mm Y 0.00 mm Z -9.00 mm
Static Video Calibration
Start
Cancel


3D Orthogonal
-X
LIVE




3D Orthogonal
-X
LIVE





Quick-Start Navigation

Previous	Quick Start Part 2/3: Camera Calibration
Next	*n/a*